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An integrated optical modulator operating at cryogenic temperatures

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Supplementary information

Supplementary Note 1 – Applications and requirements of cryogenic PICs

Cryogenic PICs have been envisioned since many years. However, few such concepts have ever been implemented, largely due to the lack of a functional technology platform in which to implement them. There are plenty of concepts and applications of cryogenic PICs which have been published, such as photonic quantum computing, superconducting supercomputing, electro-optic transduction of qubits, trapped ion quantum computing, particle detectors, and focal plane arrays. Additional applications not yet presented in scientific literature are likely to be triggered with the availability of such cryogenic photonic technology.

What all these applications have in common is a need for electro-optic switches and/or modulators as part of the photonic technology. The specific requirements on such devices varies widely between the different applications. Below we will discuss the general concept of each application of cryogenic PICs and the requirements they impose on the electro-optic switches and modulators. Today, most of these applications only exist as theoretical concepts. We therefore focus on qualitative aspects of the device requirements rather than quantitative as that is dependent on the practical implementation.

Photonic quantum computing relies on single photons as carriers of quantum states. The most promising platform to realize such a concept is silicon photonics^{S1}. While most of the components needed for photonic quantum computing can operate at room temperature, single-photon detectors rely on superconducting materials, which requires the system to operate at few-K temperatures¹. The use of single photons as qubits makes it critical to minimize the loss in every component of the PIC in order to preserve the quantum information¹⁷. To make a deterministic source of single photons, large-scale heralding is required³. In such

architectures, a fast and efficient electro-optic switch to route the heralded photons is a critical component. Because of the large number of components required to build a photonic quantum computer, the power consumption of each component is a vital aspect. Low-power phase shifters are essential to tune the PIC and to compensate for process variations¹⁷. The most important requirements for electro-optic switches in photonic quantum computing are low optical losses, low power consumption, and high electro-optic bandwidth.

Superconducting supercomputing, or rapid single flux quantum (RSFQ) computing uses quantum effects at low temperature to build efficient supercomputers based on classical computing paradigms. These computers need to transmit data to the room-temperature environment at high data rates and with low heat leakage into the cryogenic environment^{S2}. Optical interconnects are perfectly suited for this application^{S3}. Because RFSQ uses low amplitude micro-wave signals (<10 mV), directly driving an electro-optic modulator would require extremely efficient modulation. To reach more realistic driving voltages (~100 mV), cryogenic amplifiers can be used to amplify the data signal^{S4}. Even with amplification of the driving voltage, very efficient modulators are required. Because power dissipation is an important factor, also low optical losses are essential in addition to a low electric switching energy.

Electro-optic transduction of qubits is a proposed concept to convert microwave qubits to single-photons in order to preserve the quantum information for long-distance quantum communication⁷. Because microwave qubits require mK temperatures, they cannot be used to transmit quantum information to room temperature, or over larger distances, e.g. to interconnect multiple quantum systems⁷. The only viable carrier of quantum information under these conditions are photons. Electro-optic transduction is therefore needed to convert the microwave qubit to an optical qubit. The proposed devices for direct electro-optic transduction relies on the use of an optical resonator with a Pockels material coupled to a microwave resonator⁸. The conversion efficiency is directly dependent on the quality (Q) factor of the optical resonator, and on the Pockels coefficient of the material. To reach high Q -factors, low-losses are an intrinsic requirement, making that the most important parameter along with a large magnitude of the Pockels coefficient (SN 2).

Trapped ion quantum computing is one of the leading concepts for universal quantum computing based on quantum states of trapped ions. It has recently turned out^{S5} that cryogenic temperatures (4 K) are essential in order to achieve long ion lifetimes, which is key for realizing and operating systems with hundreds to many thousands of ions. One of the main technological challenges in developing such scalable platforms lies in the optical manipulation and measurement that is necessary on the level of individual ions. To this end, integrated photonics has been proposed as a scalable solution¹⁰. Because of the energy levels of the transitions in the ions, operation from the near ultraviolet to the near infrared wavelength range (above the

bandgap of silicon) is required. Initial demonstrations of the concept rely on SiN photonics with off-chip control of the light. But to scale this approach to larger number of ions where light delivery and on-chip detection using superconducting nanowires need to be controlled and multiplexed on chip^{S6}, low-power modulators and tuning elements for visible light need to be integrated with SiN waveguides. However, Pockels modulators operating at visible wavelengths have only been demonstrated at room temperature^{S7,S8}.

Particle detectors as used for example at high-energy particle colliders often need to operate at cryogenic temperatures in order to reduce electrical noise. The same holds for *focal plan arrays (FPAs)* used in radio astronomy. Like superconducting supercomputers, optical interconnects can provide low-power and low heat-leakage interconnects to transmit the recorded data to room-temperature^{11,12}. To minimize the power dissipation, low-losses and low-power operation are the most important criteria.

Beyond these examples of known applications, new opportunities might arise in new fields such as space applications, or sensing at ultra-low temperatures with the availability of a viable technology for cryogenic integrated photonics.

Supplementary Note 2 – BaTiO₃ for cryogenic interconnects

As discussed in the main text, one of the potential applications for cryogenic PICs is cryogenic interconnects. Such interconnects are needed both for transmitting classical information (e.g. from RSFQ systems⁶) and for quantum information (from quantum computers). In most cryogenic computing systems, the signal amplitude is on the order of mV, presenting a challenge in terms of driving voltage for electro-optic transmission. However, a fundamental difference exists between classical and quantum systems, in that the signals from classical systems can be electrically amplified at cryogenic temperature^{S4}, whereas amplification of quantum signals results in a loss of the quantum state.

For transmission of classical signals, cryogenic amplifiers are commercially available which can amplify mV signals to the order of ~100 mV. The high-speed modulator presented in this work used a driving signal of ~1 V, which would need to be reduced by one order of magnitude to reach the required voltage levels. A direct way to reduce the drive voltage would be to increase the Q -factor of the resonator, since the drive voltage scales inversely with the Q -factor. The current device has a $Q = 6'000$, which means $Q = 60'000$ would enable the required drive voltage, while maintaining a bandwidth in the GHz range. An alternative approach to reduce the drive voltage while maintaining a large bandwidth is to use a Mach-Zehnder modulator (MZM). At room temperature, MZMs with a bandwidth of 100 GHz and driving voltage of ~200 mV have been demonstrated using lithium niobate (LN)²⁰. The significantly larger Pockels coefficient of BaTiO₃ (200 pm/V at 4 K) is expected to enable the design of even more efficient devices.

For direct quantum-coherent electro-optic conversion of microwave qubits at cryogenic temperatures to single optical photons (quantum transduction), a microwave resonator is coupled to an optical resonator via the Pockels effect in a non-centrosymmetric material⁸. While this can be implemented using any Pockels material, for efficient conversion, a high electro-optical coupling rate is required, and this coupling rate depends linearly on the Pockels coefficient. Moreover, the optical pump power required for optimal conversion depends inversely on the square of the coupling rate and thus inversely on the square of the Pockels coefficient. The Pockels coefficient of $r \sim 200$ pm/V in BaTiO₃ at low temperature therefore provides a significant advantage over alternative materials such as LN ($r \sim 30$ pm/V at room temperature) or AlN ($r \sim 1$ pm/V at room temperature) for this specific application.

Through these improvements BaTiO₃ offers a clear path to functional cryogenic interconnects for both classical and quantum systems.

Supplementary Note 3 – Propagation losses

To investigate whether there is any significant contribution to propagation losses from the BaTiO₃ (BTO) layer we fabricated different length waveguide structures of identical dimensions with and without the BaTiO₃ layer present. The non-distinguishable (within error) propagation losses in both structures (Figure S1a) indicates that the propagation losses of our devices are limited by our SiN waveguide fabrication process. This conclusion is corroborated by the fact that losses are lower in the Si-BaTiO₃ waveguides which uses a fabrication process-flow that has undergone greater optimisation in our cleanroom. Since SiN waveguides with much lower propagation losses can in principle be manufactured^{S9}, the SiN-BaTiO₃ waveguide losses are expected to be improved significantly with an improved fabrication process.

The fixed loss per unit length of the waveguide structures leads us to believe that sidewall roughness is the dominant cause of loss in our devices. It is expected therefore that the loss should be temperature independent. To verify this we measured the propagation loss of SiN-BaTiO₃ devices at 4 K and compared the device results with room temperature (300 K) and found no distinguishable (within error) difference (Figure S1b).

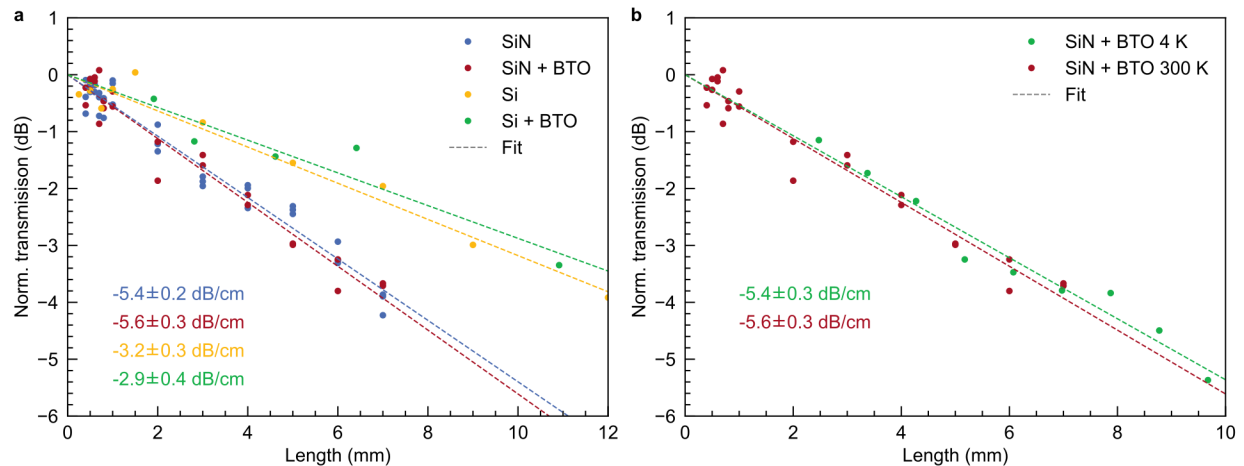


Figure S1. Comparison of waveguide losses. Cut-back loss-measurements of **a** SiN and SiN-BaTiO₃ as well as Si and Si-BaTiO₃ waveguides showing no measurable losses caused by BaTiO₃, and **b** SiN-BaTiO₃ waveguides at 300 K and 4 K showing a change within measurement error in losses with temperature.

Using the measured losses of simple waveguide structures, we can estimate the insertion loss of the 2x2 switch reported in the manuscript. Accounting for coupling loss, the 2x2 Mach-Zehnder interferometer exhibits an on-chip insertion loss of 1.6 dB. The 2-mm-long access waveguides contribute 1.1 dB of propagation loss resulting in an insertion loss of the switch of 0.5 dB. Variation in the coupling efficiency caused by manual fibre alignment and limited mechanical stability of the cryogenic probe station introduced

some degree of uncertainty to this value. However, considering the waveguide length of the switch ($\sim 600 \mu\text{m}$), the negligible electrode loss (ensured by the large electrode gap), and the low-loss nature of multi-mode interference splitter, an insertion loss of $<1 \text{ dB}$ is estimated.

In the case of the resonant modulator used for data modulation, the losses can be analysed based on the loss and coupling coefficients (α, t) of the resonator^{S10}:

$$(\alpha, t) = \left(\frac{A}{B}\right)^{\frac{1}{2}} \pm \left(\frac{A}{B} - A\right)^{\frac{1}{2}}$$

$$A = \frac{\cos \frac{\pi}{F}}{1 + \sin \frac{\pi}{F}}$$

$$B = 1 - \left(\frac{1 - \cos \frac{\pi}{F}}{1 + \cos \frac{\pi}{F}}\right) \frac{1}{\text{ER}}$$

$$F = \frac{\text{FSR}}{\text{FWHM}}$$

Where ER, FSR, and FWHM are the extinction ratio, free spectral range and full width at half maximum, respectively, of the resonator. Using the parameters measured for the resonant modulator (**Figure S2**) results in loss and coupling coefficients $\alpha, t = 0.92$ ($\alpha = t$ when critically coupled). When combined with the circumference of the resonator this value results in a propagation loss of 23 dB/cm . This is significantly higher than the access waveguide loss of 2.9 dB/cm (**Figure S1**) due to the close proximity of the electrodes to the waveguide.

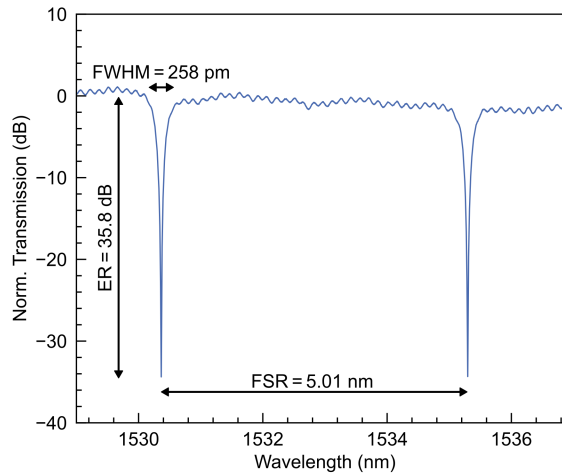


Figure S2. Normalised transmission spectrum of the resonant modulator used for high-speed data modulation experiment. The annotations indicate the determined extinction ratio (ER), full width at half max (FWHM) and free spectral range (FSR) of the resonator.

From the coupling and loss coefficients it is also possible to calculate the off-resonance insertion loss of the modulator^{S11}:

$$\frac{(\alpha + t)^2}{(1 + \alpha t)^2} = 0.993 = -0.3 \text{ dB}$$

Supplementary Note 4 – Determination of r_{eff}

We extract the effective Pockels coefficient of BaTiO₃ from the measurements of nonlinear optical hysteresis in Δn_{BTO} as a function of electric field. Because domain switching effects contribute to the NLO response in the DC regime (resulting in the hysteresis observed in Figure 2a), extraction of the Pockels coefficient must take such effects into account. The ferroelectric hysteresis of BaTiO₃ is directly related to Δn_{BTO} through the fraction of poled domains, ν , according to:

$$\Delta n_{\text{BTO}} = -\frac{1}{2}r_{\text{eff}}\nu n_{0,\text{BTO}}^3 E \quad (1)$$

where E is the applied electric field, and $n_{0,\text{BTO}}$ is the refractive index of BaTiO₃ with no field applied. The poling fraction of a ferroelectric material can be described by^{S12}:

$$\nu = \frac{\sinh(A(E \pm E_C))}{A/B - 1 + \cosh(A(E \pm E_C))} \quad (2)$$

where E_C is the coercive field, A and B are material parameters^{S12}. Inserting the expression for ν in eq. (1) we can determine the material parameters in eq. (2) as well as r_{eff} by fitting to the measured Δn_{BTO} . To model the optical response at large electric fields, we introduced a parameter, k , for the third-order nonlinear optical response:

$$\Delta n_{\text{BTO}} = -\frac{1}{2}r_{\text{eff}}\nu n_{0,\text{BTO}}^3 E + kE^2 \quad (3)$$

In this work we derived the Pockels coefficients from our data using eq. (3) as the fit. An example of the quality of the fit is shown in Figure S3; the fitting parameters of which are given in Table S1. By including the ferroelectric hysteresis in the evaluation, we can separate the contributions from domain switching and the Pockels effect. This ensures that the true r_{eff} can be extracted, separate from domain switching effects.

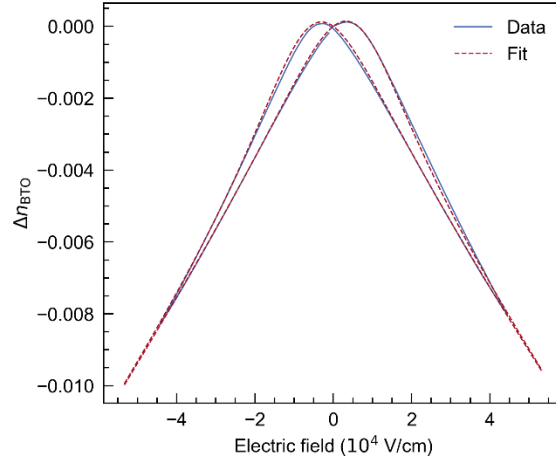


Figure S3. NLO hysteresis measured in a BaTiO₃-SiN device at 300 K and the fitted model based on equation (3).

Table S1. Parameters for the fit showed in Figure S3. $n_{0,\text{BTO}}$ was fixed to 2.29 based on ellipsometry measurements at room temperature.

r_{eff} (pm/V)	E_C (10^6 V/m)	A (10^{-6} m/V)	B (10^{-6} V/m)	k (10^{-18} m ² /V ²)
321	0.606	1.09	0.762	16.7

Supplementary Note 5 – Electrical characterisation

Electrical I-V and C-V measurements were performed on dedicated test structures consisting of 2-mm-long electrodes separated by 1.25 μm (C-V) or 2 μm (I-V) which were patterned on top of 80-nm-thick BaTiO₃. The test structures were mounted in a cryogenic probe station and characterised at temperatures in the range 4 K to 300 K. At 4 K, the current measured in the I-V sampling is dominated by capacitive charging and ferroelectric switching. Above 150 K, thermally activated leakage current starts to dominate (Figure S4). The leakage mechanism in titanate perovskites such as BaTiO₃ can typically be described by a modified Schottky emission equation^{S13} which describes a thermally activated charge transport. Indeed, we observe such thermal activation for temperatures above 150 K. Below that temperature, the experimentally measured leakage is linked to ferroelectric switching currents and capacitive charging.

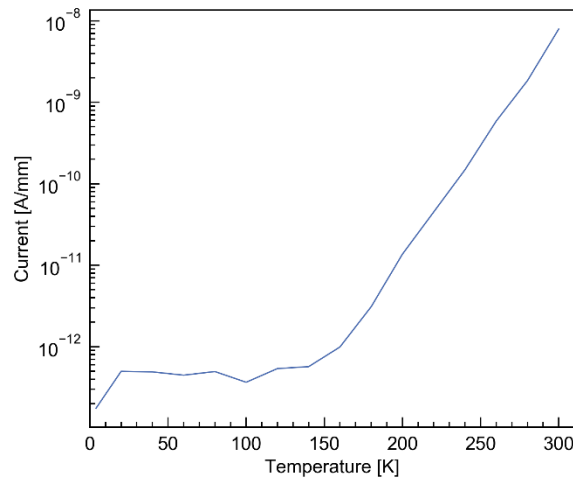


Figure S4. Leakage current through BaTiO₃ at various temperatures showing a reduction by more than 10⁴ from 300 K to 4 K.

The different contributions to the measured current and their relation to ferroelectric switching are illustrated in Figure S5. ① When the applied field is at a maximum the material domains and dielectric are fully polarised and the current flowing is only that associated with leakage. ② When the applied field is reduced to zero (a fraction of) the ferroelectric domains alone remain polarised. ③ When a small opposing field is applied the ferroelectric domains begin to switch their polarisation, the dielectric is also charging and these combine to form a current maximum. ④ The material is now in the reversed state to ① and the current is dominated by a leakage current flowing in the reverse direction.

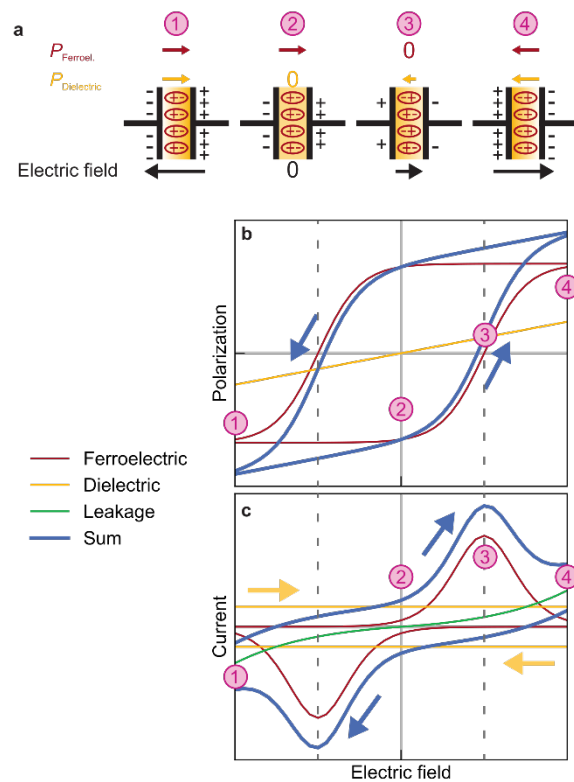


Figure S5. Change in polarisation and resulting electrical current in BaTiO_3 capacitors. a Polarisation states in ferroelectric capacitor at different field strengths. b The calculated polarisation and, c, current showing contributions from ferroelectric domain switching, dielectric charging, and leakage. The numbered points 1-4 are described in the text above.

Supplementary Note 6 – Field induced phase transition

As described in the main manuscript, the electrical and electro-optic data suggests the presence of a field-induced phase transition in BaTiO₃ at temperatures below ~100 K (Figure 3). At temperatures below 100 K, the electro-optic response of the device changes sign when the electric-field strength exceeds 10-15×10⁴ V/cm (Figure S6). This change indicates the appearance of negative elements in the Pockels tensor, which can only be caused by a change in the crystal symmetry – a phase transition.

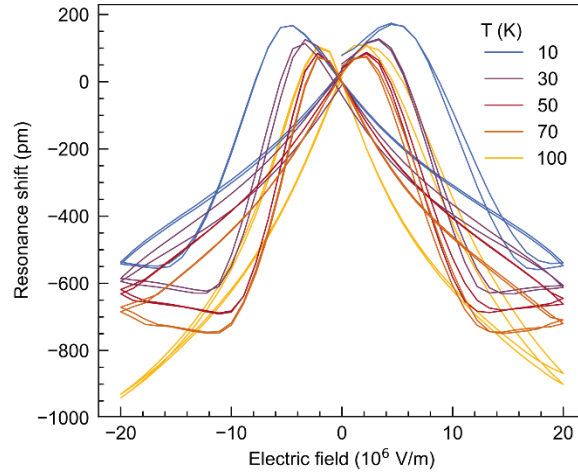


Figure S6. Optical hysteresis curves at temperatures below 100 K. Induced resonance shifts at various temperatures from 10 to 100 K in a device oriented along the 45° direction (BaTiO₃[011]) showing evidence of a field-induced phase transition below 100 K at an electric field strength of 10-15×10⁴ V/cm.

Indeed, in the case of the phase transitions of bulk BaTiO₃ the low-temperature transition from orthorhombic (space group Pmn2₁) to rhombohedral (space group R3m) structure includes the appearance of negative elements (r_{22})^{S14}:

$$\begin{pmatrix} 0 & 0 & r_{13} \\ 0 & 0 & r_{23} \\ 0 & 0 & r_{33} \\ 0 & r_{42} & 0 \\ r_{51} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}_{Pmn2_1} \Rightarrow \begin{pmatrix} 0 & -r_{22} & r_{13} \\ 0 & r_{22} & r_{13} \\ 0 & 0 & r_{33} \\ 0 & r_{42} & 0 \\ r_{42} & 0 & 0 \\ -r_{22} & 0 & 0 \end{pmatrix}_{R3m}$$

A transition to a rhombohedral phase at large electric fields, something not observed in bulk BaTiO₃, could trigger the change in the electro-optic behaviour as described above.

Phase transitions are also expected to affect the dielectric properties of a material and we expect to see a peak or kink in permittivity around the phase transition temperature. Because of the micro-domain structure of BaTiO₃ thin-films, such peaks can be significantly broadened. C-V data (as shown in Figure 2e) acquired

at various temperatures was used to extract the permittivity of the BaTiO₃ layer. The permittivity was extracted by performing electro-static simulations where the BaTiO₃ permittivity was varied to match the measured capacitance as described in more detail in the supplementary information of ref. 3.

Indeed, we observe a kink in the permittivity at large bias fields around 100 K (Figure S7). We can also see that there is a peak around 240 K, corresponding to the phase transition observed in the temperature dependent EO response (Figure 3)

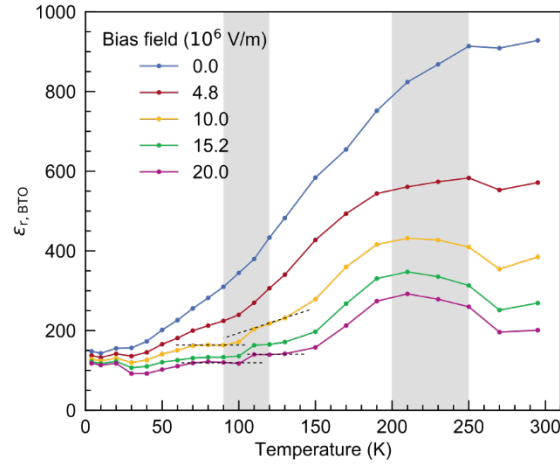


Figure S7. Temperature dependence of BaTiO₃ permittivity at different electric-field strengths along the <100> (0°) direction. A clear peak around 240 K is observed for all field-strengths, and at high field strengths a kink appears around 100 K, matching the phase transitions observed in the electro-optic response (Figure 3). The kinks in the 10 and 20 MV/cm curves are indicated by dashed lines. The grey areas indicate the temperature regions of the two phase-transitions.

Supplementary Note 7 – Switching energy

The Mach-Zehnder interferometer (MZI) used for switching (Figure 4a,b) consists of 500 μm long electrodes separated by 9 μm . Mapping our experimentally determined capacitance values (Figure 2e) to the geometry of the MZI results in a total device capacitance of ~ 11 fF. For a full π phase shift ± 50 V are applied to the device. The switching energy of a capacitor is given by

$$E = \frac{1}{2} CV^2$$

When applying a voltage swing from $-V$ to $+V$ the total switching energy is given by

$$2 \frac{1}{2} CV^2 = CV^2$$

resulting in a switching energy of 28 pJ. Optimisation of the device geometry using thicker BaTiO₃ (240 nm) to increase the optical overlap (~ 55 %), a smaller electrode gap (4.5 μm), and a longer phase shifter (1 mm) would reduce the voltage by a factor 12 while simultaneously increasing the capacitance by a factor 12. These changes would be expected to result in a reduction of the switching energy to only 2.3 pJ.

Highly-doped p-n junctions cannot be used for low-loss switching because of the large propagation losses and the simultaneous modulation of the (real part of the) refractive index and the absorption. Thermo-optic switches do not have such optical loss issues and are better suited for switching. However, in thermo-optic switches the power consumption is dominated by the static current through the device. It is therefore necessary to specify the time needed for switching the state of the device. Due to the low bandwidth, thermo-optic devices are typically limited to ~ 1 μs switching speed. Switching at this time scale corresponds to the lowest switching energy that can be achieved. During that switching period the switch will require ~ 0.25 μJ , assuming $P_\pi = 0.5$ W¹⁴ and 50 % of the maximum current as average current applied during the switching operation. The switching energy of 0.25 μJ is $\sim 10^4$ times larger than the value we report for BaTiO₃ device, and 10^5 times larger than the value expected for an optimised BaTiO₃ device. Please note that in addition to this dynamic switching energy, also the static power consumption needed to hold the state requires significant power in thermo-optic devices (~ 0.5 W) and is almost negligible (< 10 pW) in BaTiO₃-based switches.

Supplementary Note 8 – Data modulation experiment

The cryogenic probe station used in the experiments is equipped with dual RF connections for sending a signal to a GSGSG RF probe mounted inside the system. In order to account for the impact of this connection on the electro-optical measurements, we carefully characterised the electrical performance of the RF connectors both by measuring the frequency response, and by sending data signals. We used a dedicated sample with metal lines similar to those used in the EO devices to connect the signal lines of the GSGSG probe inside the cryogenic probe station. The S_{21} parameter in that configuration show an attenuation of 10-20 dB in the frequency range of 10-30 GHz when the signal is transmitted through the cryogenic probe station (Figure S8). We compensate for the attenuation in the S -parameter measurements of the photonic devices (Figure 2c) by calibrating the setup using a calibration kit. For data transmission measurements (see below) such calibration is not possible.

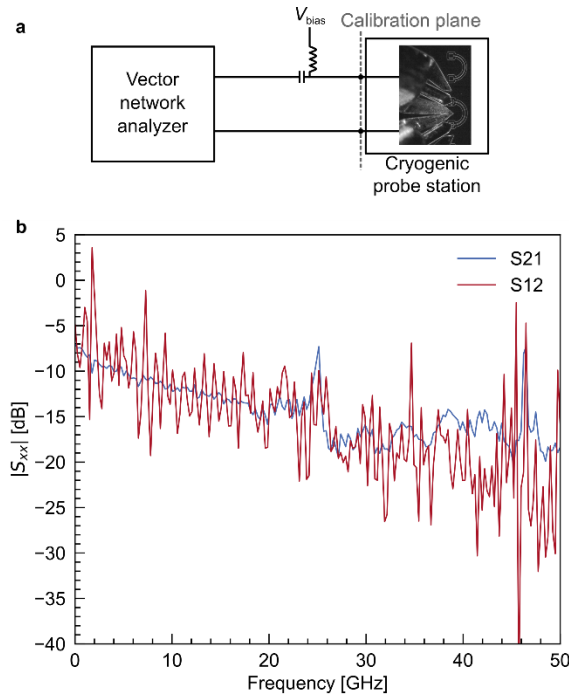


Figure S8. S-parameter measurements of the transmission through the cryogenic probe station. **a**, the experimental configuration used to perform the measurements. The calibration plane was adjusted to the connections to the cryostat. **b**, the measured transmission S -parameters through the cryostat. We observe large frequency dependent losses with 10-15 dB attenuation in the 10-20 GHz range.

To evaluate the performance of the BaTiO₃ photonic modulator, we analysed the signal integrity and estimated the actual voltage applied to the electrodes of the modulator when performing data modulation experiments.

We also characterized the frequency response of the modulator, using electro-optic S -parameter measurements (Figure S9), to ensure its suitability for high-speed operation. The S_{21} -response shows a cut-

off at 30 GHz which corresponds very well with the expected bandwidth limitation of 32 GHz caused by the Q -factor of $\sim 6'000$.

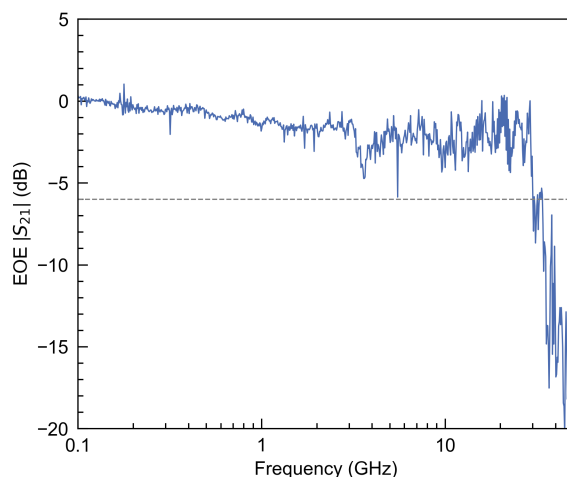


Figure S9. Electro-optic S_{21} -parameter measurements of the modulator used for data transmission experiments. The cut-off at ~ 30 GHz (indicated by dashed line) is caused by the photon lifetime in the resonator and matches very well with the theoretical value (32 GHz) based on the measured Q -factor of $\sim 6'000$.

As discussed above, RF losses in the setup severely impact the electrical signal applied to the electrodes in the sample. We compensated for part of bandwidth limitations in the electrical path by pre-distorting the electrical signal generated in an arbitrary waveform generator (AWG) (Figure S10a). However, the eye diagram (Figure S10a) still shows significant noise and a limited time-response which indicates the limitations to our electro-optical characterisation setup.

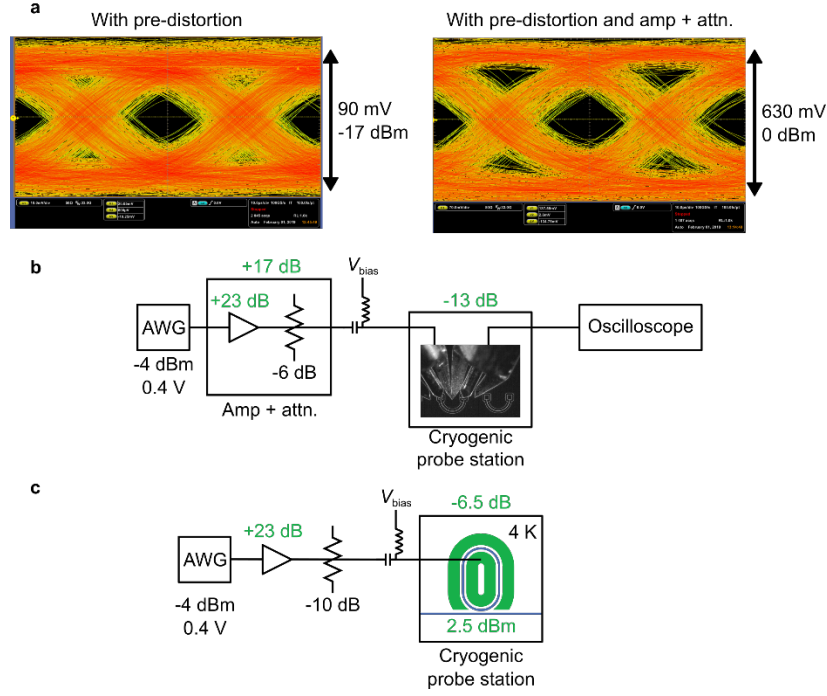


Figure S10. Characterisation of modulation voltage in cryogenic probe station. **a**, by measuring the signal amplitude after being sent through the cryogenic probe station, with and without amplification we can extract the contribution of the individual components in the system. **b**, we find that the RF amplifier provides 23 dB of gain, and that transmitting the signal through the cryogenic probe station an attenuation of -13 dB. **c**, we can extract the real drive voltage based on the values for the RF amplifier and the cryogenic probes station. Assuming equal losses in to and out of the probe station results in 6.5 dB attenuation of the RF signal at the device. Combining all gain/attenuation values results in 2.5 dBm RF power on the device, which corresponds to $\sim 1.7 V_{pp}$ on an open termination.

Besides the electrical noise in the drive signal, the electro-optic eye-diagram shows additional high-frequency noise. This additional noise is a consequence of low optical input power (due to low coupling efficiency in the cryogenic probe station) on the optical amplifier used in the experiment (schematic in Figure 4c). Because of the low input power, the optical amplification results in additional noise. This noise is unrelated to the modulated signal as is evidenced by the large high-frequency noise also at lower bit rates (Figure S11). This limitation to our experimental setup results in the observed degradation of the electro-optic eye diagram but is not related to the modulator performance.

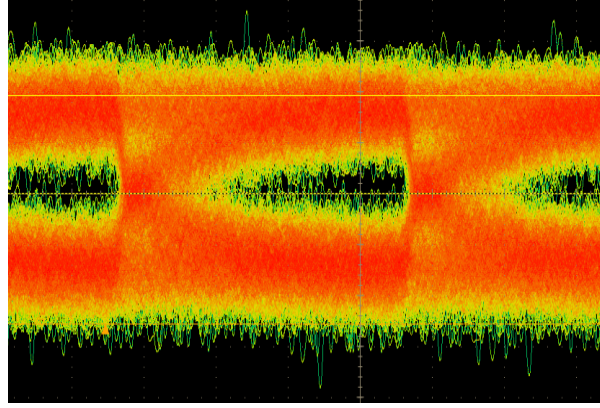


Figure S11. Electro-optic eye diagram at 1 Gbps showing the same high-frequency noise as the eye diagram recorded at 20 Gbps (Figure 4d). This eye diagram was recorded without pre-emphasis to compensate for the electrical bandwidth limitation.

By comparing the magnitude of the electrical signal transmitted through the cryogenic probe station (as described in Figure S10), with and without amplification, to the magnitude of the source output, we could determine the contributions of the individual components in the setup (Figure S10b). Based on the contributions of the individual components in the setup, we can extract the real drive voltage applied to the device during the electro-optic data modulation experiments. We estimate the drive voltage applied to the EO modulator to be $V_{pp} \sim 1.7$ V (Figure S10c). Using an estimated device capacitance of 62 fF (based on an identical sample stack used in ref. 3 and assuming the same reduction of permittivity with temperature as observed in this work), we determine the modulation energy to be 45 fJ/bit according to

$$E_{bit} = \frac{1}{4} CV^2$$

Supplementary Note 9 – Coercive field of BaTiO₃

The temperature dependent electro-optic measurements allow us to extract the effective coercive field of BaTiO₃ as a function of temperature (Figure S12). The effective coercive field is the measured orientation-dependent coercive field, which scales from the intrinsic coercive field of BaTiO₃ depending on the orientation of the applied electric field (because of the domain structure of BaTiO₃). The effective coercive field along the 90° direction is equal to the intrinsic coercive field.

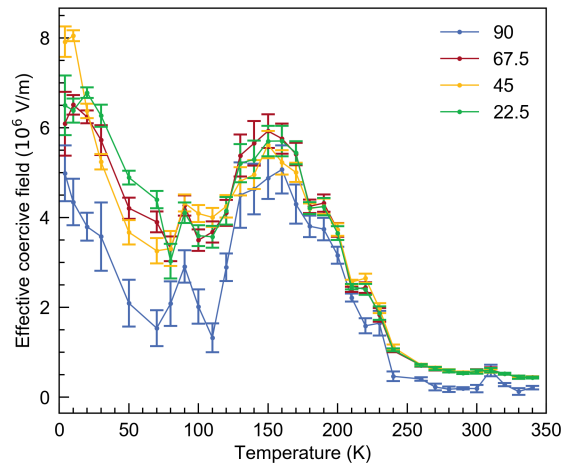


Figure S12. Temperature dependence of the effective coercive field of BaTiO₃ extracted from electro-optic measurements (SN 3).

We observe a general trend that the coercive field increases with decreasing temperature. In addition, we observe a sharp drop in the coercive field in the same temperature range where the field-induced phase transition appears (SN 6). Along the direction of the most efficient electro-optic modulation (45°), the effective coercive field at 4 K is $\sim 8 \times 10^6$ V/m.

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